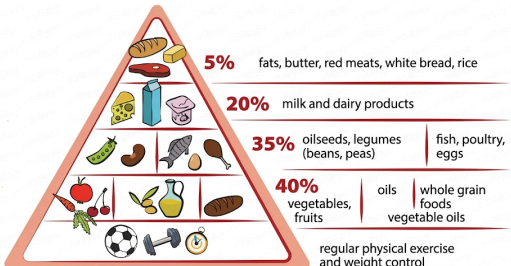


TOPIC 5.A

The human digestive system and healthy nutrition

1. What are the midgut (small intestine) sections of the digestive system and what is their function?
2. How would you illustrate the nutrients of a healthy diet, their quantities, and their proportions relative to each other?
3. What are these nutrients?
4. What is the role of water and minerals in our diet?

Expected knowledge	Attainable score	Attained score
1. • The first section of the midgut is the duodenum. The contents of the stomach, bile, and pancreatic juice are emptied here. • The 3 sections of the small intestine: duodenum, jejunum, ileum. • Bile is produced in the liver and stored in the gallbladder. Its function is the emulsification of fats. Its composition: red blood cells [breakdown products], bile acids. • Pancreatic juice: produced in the pancreas (an organ with dual secretion). Its enzymes: trypsin, lipase, amylase, nucleases. • Intestinal juice: its production, enzymes. • Structure of the small intestine: intestinal villi, glove-finger-like. • Absorption through its epithelial cells.	14 points	
2. & 3. 2-3.  • Food pyramid / Healthy eating plate. • Nutrients: proteins, carbohydrates, fats, vitamins, minerals, water.	5 points	
4. • Water: solvent, transport medium, reaction medium, etc. • Potassium: muscle work, regulation of	3 points 3 points	

fluid balance. • Iron: component of haemoglobin.		
5. Structure of the answer, communication skills	5 points	
Total:	30 points	

TOPIC 5.B

Characterisation of a plant species

1. Identify the received plant species using the Plant Identification book. Describe the process of identification.
2. To which major taxonomic groups does it belong? What are the characteristics of these groups that can be found on the received species?
3. Characterise the ecological requirements of the species.
4. In what type of habitat is the occurrence of this species typical? Where can we find such a place around the school?



Expected knowledge	Attainable score
1. Naming the species, listing the identification keys. (Elderberry - <i>Sambucus nigra</i>)	5 points
2. Taxonomic group: phylum Angiosperms, class Dicotyledons • pinnate leaf venation, taproot system, • compound leaves, dentate (serrate) leaf margin.	5 points
3. Listing ecological requirements: moist, mesic (fresh) area; nutrient-rich soil; nitrogen indicator plant.	5 points
4. Around the school, it occurs everywhere next to the fence in neglected areas and forest edges. Medicinal plant: its inflorescence is used for diaphoretic tea (to induce sweating), and its berries are rich in vitamins.	5 points
Total:	20 points